Fraud Lens: An AI-Powered System for Detecting and Preventing E-Commerce Refund Abuse

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Abstract

- Fraud Lens uses AI and machine learning to detect refund fraud in e-commerce.
- Analyzes large transaction data in real time to identify suspicious patterns.
- Generates automated fraud reports using NLP for faster investigation.
- Provides dashboards and collaboration tools for better fraud monitoring and decision-making.
- Reduces financial losses and detection time by automating fraud monitoring.

Problem Statement and Objectives

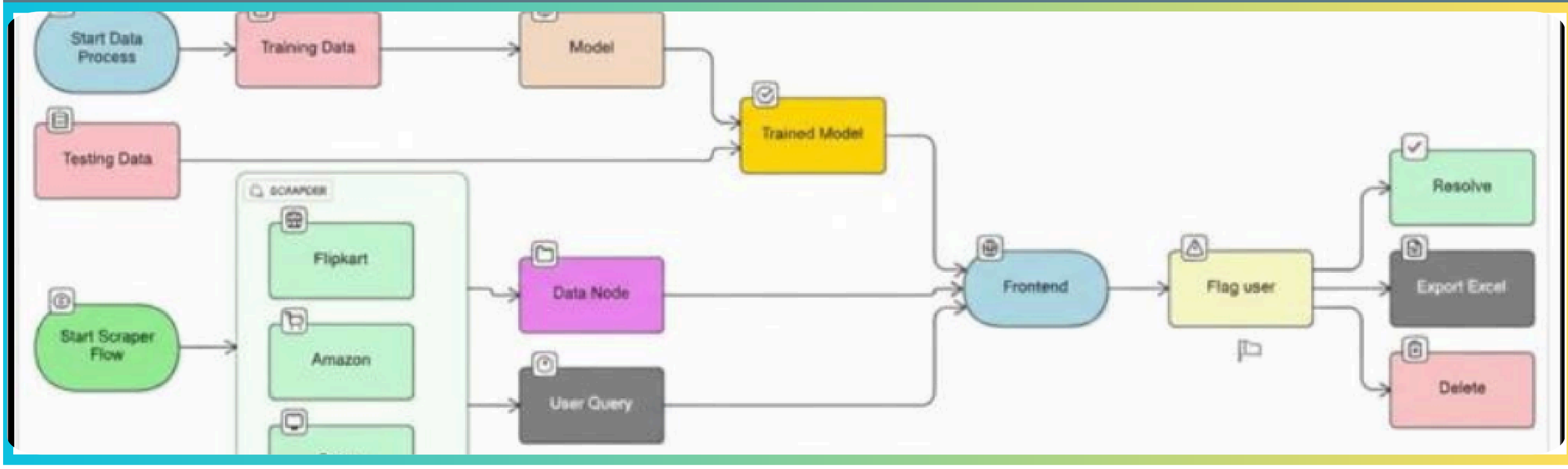
Problem Statement

- Increasing fake refund claims cause major financial losses for e-commerce platforms.
- Manual reviews and rule-based systems are slow and often miss new fraud patterns.
- Existing systems struggle to quickly identify suspicious behavior in large transaction datasets.

Objectives

- Automate refund fraud detection using AI and machine learning.
- Detect fraud in real time and reduce financial losses.
- Provide analytics dashboards for fraud monitoring and insights.
- Integrate securely with e-commerce systems while ensuring data privacy.
- Use adaptive learning models to continuously improve fraud detection accuracy.

System Design



Results

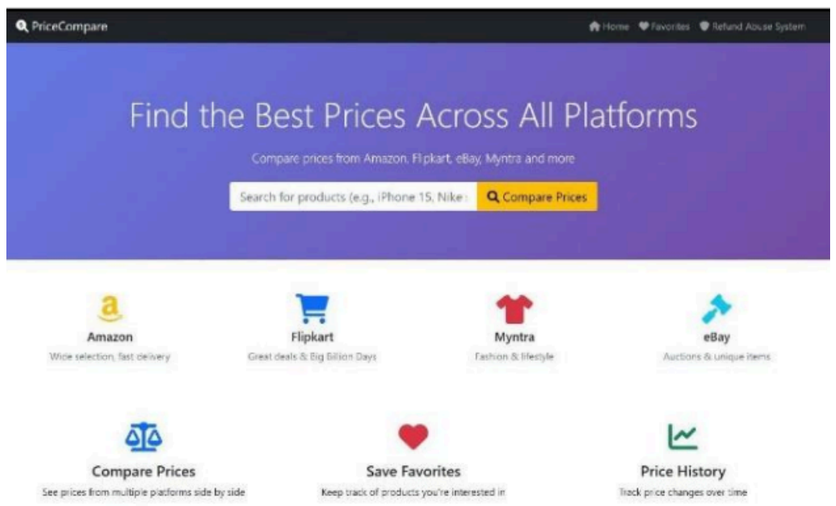


Fig 1. Centralized Navigation Hub

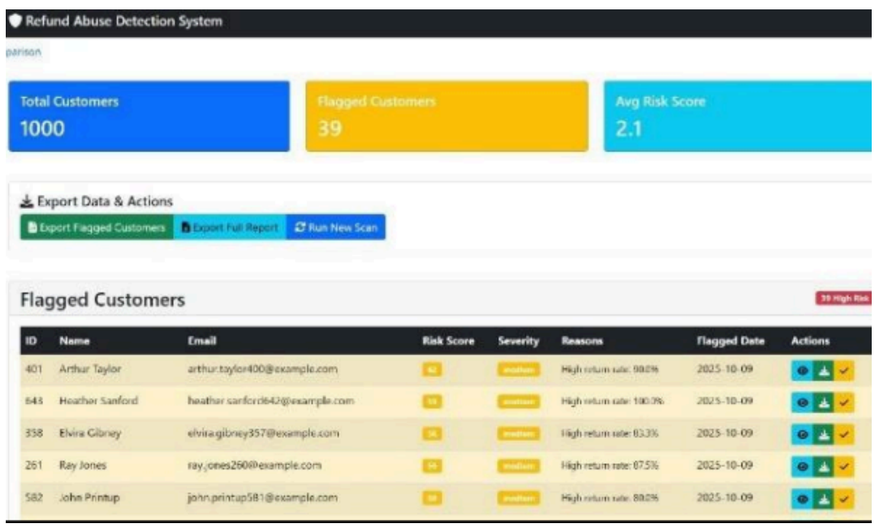


Fig. 2. Refund Abuse Detection System Dashboard

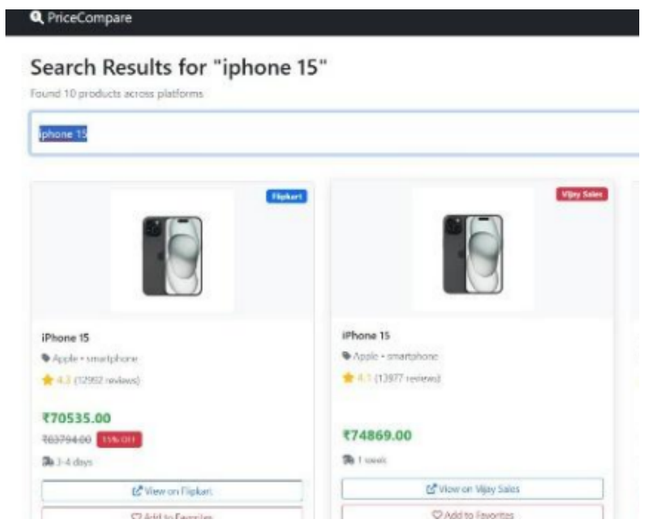


Fig. 3. Price Compare Search Landing Page

SDG Goals

Conclusion

- Monitor user behavior in real time to detect refund fraud before it occurs.
- Combine pattern recognition and subtle anomaly detection for smarter fraud spotting.
- Use AI tools like computer vision and language models to verify returns and analyze suspicious behavior.
- Enhances trust and protects profit by reducing fraud while keeping legitimate customers friction-free.

References

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[6] Giriraj Daga, Reshma Shrivastava, 2023 – Return to Origin Fraud in the E-Commerce Industry – A Comprehensive Analysis (directly analyzes refund fraud patterns).

[7] Priyanka Shiwarkar et al., 2022 – A Fraud Return Detection System in E-commerce (presents a system specifically for detecting fraudulent returns).